Akoizo Original Practice - 2027 Rotation Form 3 - Answer key

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) An LED’s forward voltage is…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Always zero

B. Equal to any source

C. Device- and condition-dependent

D. Always 0.7 V

Correct: C

2. (2 points) A watt-hour measures…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Resistance

B. Charge alone

C. Energy

D. Current

Correct: C

3. (2 points) Negative solved current means…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. No voltage exists

B. Opposite the assumed direction

C. Impossible circuit necessarily

D. Negative charge disappears

Correct: B

4. (2 points) A diode is generally…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. A mechanical switch

B. Nonlinear

C. An ideal fixed resistor

D. Always conducting both ways

Correct: B

5. (4 points) Why is charge not consumed?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Steady charge flow conserves charge while energy changes form. (4 points).

6. (4 points) Write a source-two-resistor loop equation.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: Vs - IR_1 - IR_2 = 0 for traversal with current. (4 points).

7. (4 points) Name two LED faults.

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Reversed orientation and an open path; node measurements can distinguish them. (4 points).

8. (4 points) Why is Ah not sufficient for battery energy?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Voltage and operating conditions are also required. (4 points).

9. (4 points) Explain a nodal branch current.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: Current from node a to b through R is (Va - Vb)/R. (4 points).

10. (4 points) Why not infer brightness from current alone?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Efficiency, optics, temperature, and device characteristics affect light output. (4 points).

11. (2 points) A 9 V ideal battery drives a 4.5 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 5 and 7 ohms. Find equivalent resistance.

answer: 12 ohms (2 points).

Numerical target 12; absolute tolerance 0.024. Accepted units: , ohms.

13. (2 points) Two identical 10 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 5 ohms (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 9 V. Find power.

answer: 18 W (2 points).

Numerical target 18; absolute tolerance 0.036. Accepted units: , W.

15. (2 points) A node has incoming currents 6 A and 2 A and one outgoing current. Find the outgoing current.

answer: 8 A (2 points).

Numerical target 8; absolute tolerance 0.016. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

answer: In the ideal model the reverse-biased diode blocks current; a forward-biased conducting diode permits conventional current from anode toward cathode. Real leakage and forward voltage require a stated device model. (2 points).

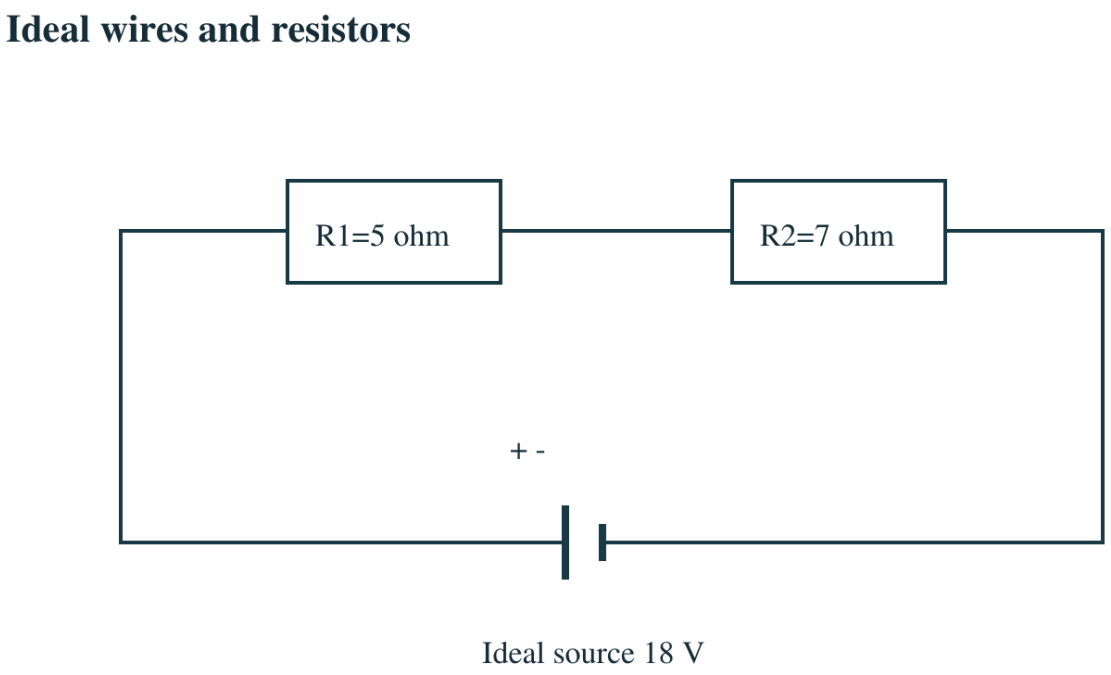
19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).

21. (2 points) Use Figure F1. Calculate equivalent resistance.



F1. Series resistor circuit (original synthetic teaching figure)

answer: 12 ohms (2 points).

Numerical target 12; absolute tolerance 0.036. Accepted units: , ohms.

22. (2 points) Use Figure F1. Calculate loop current.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1.5 A (2 points).

Numerical target 1.5; absolute tolerance 0.0045. Accepted units: , A.

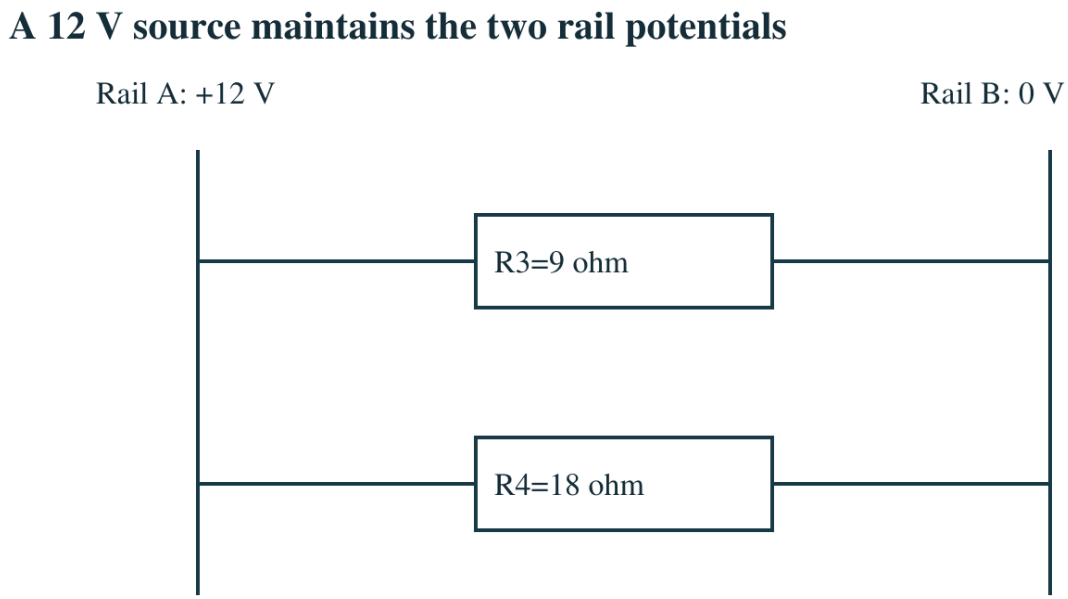
23. (2 points) Use Figure F1. Calculate the potential difference across R1.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

answer: 7.5 V (2 points).

Numerical target 7.5; absolute tolerance 0.0225. Accepted units: , V.

24. (2 points) Use Figure F2. Find current through R3.



F2. Two parallel branches (original synthetic teaching figure)

answer: 1.3333333 A (2 points).

Numerical target 1.3333333; absolute tolerance 0.004. Accepted units: , A.

25. (2 points) Use Figure F2. Find current through R4.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.66666667 A (2 points).

Numerical target 0.66666667; absolute tolerance 0.002. Accepted units: , A.

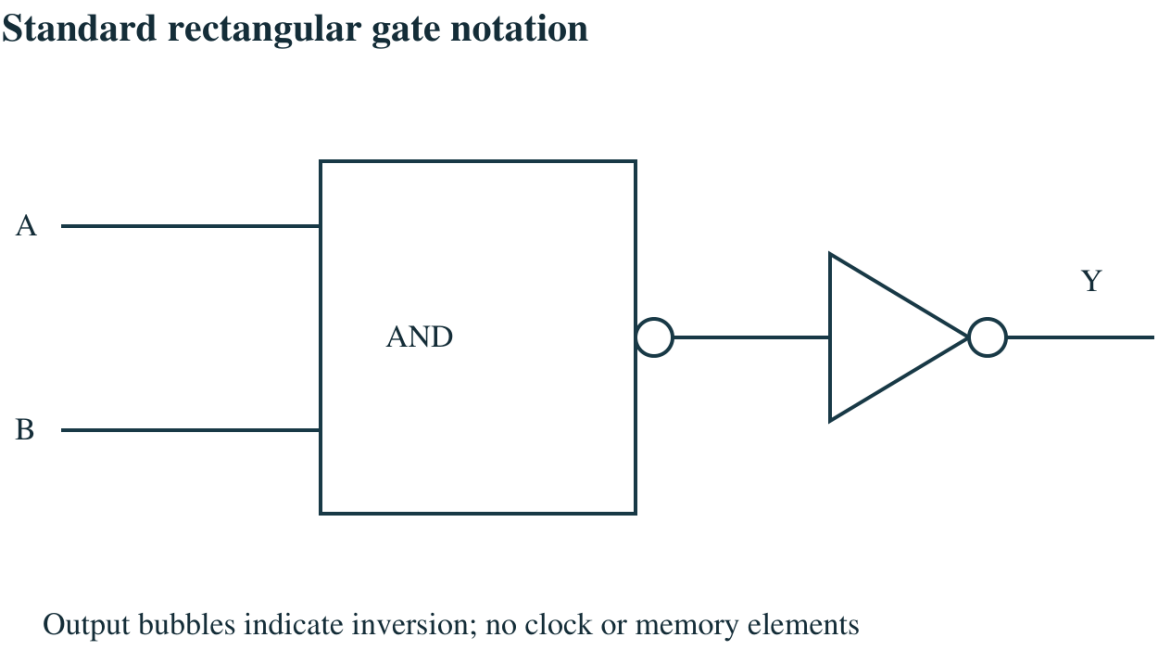
26. (2 points) Use Figure F2. Find equivalent resistance between the rails.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

answer: 6 ohms (2 points).

Numerical target 6; absolute tolerance 0.018. Accepted units: , ohms.

27. (2 points) Use Figure F3. Find Y for A=1 and B=0.



F3. Combinational logic (original synthetic teaching figure)

answer: 0 (2 points).

Numerical target 0; absolute tolerance 1e-10. Accepted units: .

28. (2 points) Use Figure F3. Find Y for A=1 and B=1.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1 (2 points).

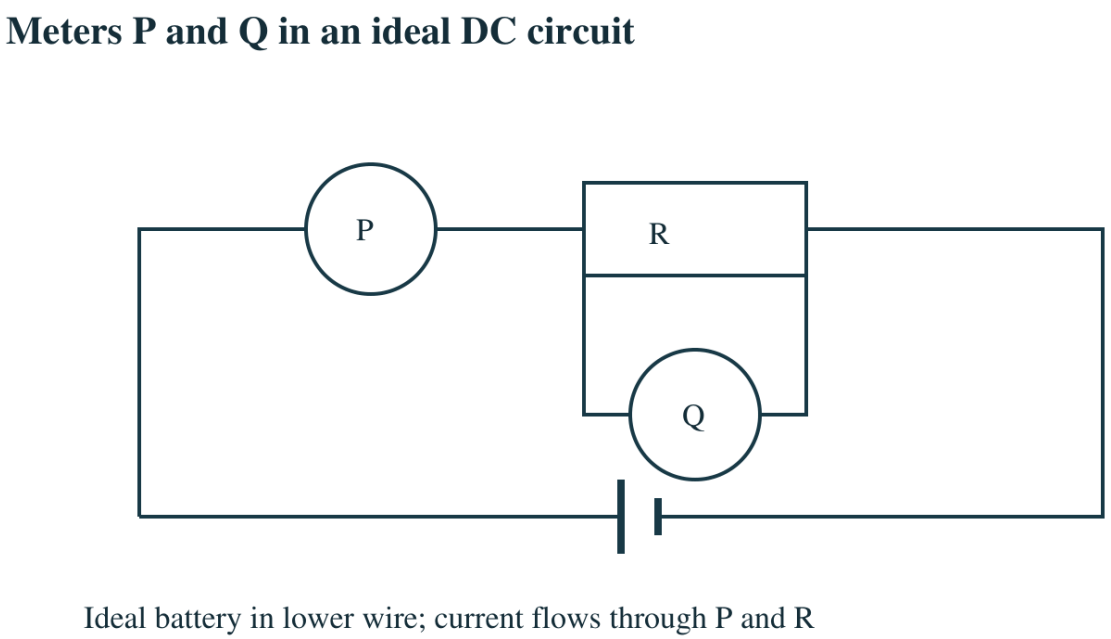
Numerical target 1; absolute tolerance 0.003. Accepted units: .

29. (2 points) Use Figure F3. Name the single gate equivalent to the complete diagram and justify.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

answer: AND. The NAND result is inverted again, so the two inversions cancel and Y=A AND B. (2 points).

30. (2 points) Use Figure F4. Which meter position is appropriate for a voltmeter? Explain using the wiring.



F4. Measurement connections (original synthetic teaching figure)

answer: Q, because it spans both terminals of R and measures potential difference in parallel. (2 points).

31. (2 points) Use Figure F4. Which meter position is appropriate for an ammeter? Explain.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

answer: P, because it lies in the series current path through R. (2 points).

32. (2 points) Use Figure F4. Describe the desired input resistance of the meter at Q and its purpose.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

answer: Very high input resistance, to draw little additional current and reduce loading of the measured circuit. (2 points).